Amitayush Thakur

amitayusht@gmail.com | in/amitayush-thakur | amit9oct | iamthakur.com

Research Interests

Automated reasoning, verified code generation, Neuro-symbolic AI, training LLMs for planning and reasoning tasks, building agentic systems

Education

University of Texas at Austin — Austin, Texas
Ph.D. Computer Science — *Aug. 2022 – Present*
Advisor: Prof. Swarat Chaudhuri

Birla Institute Of Technology and Science, Pilani — Pilani, India
Master of Science (Hons.) Mathematics — *Aug. 2012 – July 2017*
Bachelor of Engineering (Hons.) Computer Science

Publications & PrePrints

Z. Li, J. Liu, M. Vukelic, N. Shen, S. Lall, **A. Thakur**, A. Zhang, O. Khattab, J. Light, A. Solar-Lezama, ***"Harness as a Language: A Minimalist Agent Framework With Maximal Expressivity,"*** PrePrint, 2026. (Top 2% of submissions & Oral Presentation at Meta-Agents Workshop NeurIPS 2026) [arXiv][Code]

G. Tsoukalas, R. Saha, **A. Thakur**, S. Reguyal, S. Chaudhuri, ***"Learning Interestingness in Automated Mathematical Theory Formation,"*** NeurIPS, 2025. (Spotlight presentation) [arXiv][Code]

A. Thakur, J. Lee, G. Tsoukalas, M. Sistla, M. Zhao, S. Zetzsche, G. Durrett, Y. Yue, and S. Chaudhuri, ***"CLEVER: A Curated Benchmark for Formally Verified Code Generation,"*** NeurIPS, 2025. [arXiv][Code]

A. Thakur, G. Tsoukalas, G. Durrett and S. Chaudhuri, ***"ProofWala: A Framework for Multilingual Proof Data Synthesis and Theorem-Proving,"*** 17th International Conference on Interactive Theorem Proving (ITP 2026), LIPIcs, vol. 382, 2026, pp. 24:1-24:21, doi: 10.4230/LIPIcs.ITP.2026.24. [Paper][arXiv][Code]

G. Tsoukalas, J. Lee, J. Jennings, J. Xin, M. Ding, M. Jennings, **A. Thakur**, and S. Chaudhuri. ***"PutnamBench: Evaluating Neural Theorem-Provers on the Putnam Mathematical Competition."*** Neural Information Processing Systems (NeurIPS), 2024.(Best Paper Award at AI for Math Workshop ICML 2024) [arXiv][Code]

A. Thakur, G. Tsoukalas, Y. Wen, J. Xin, and S. Chaudhuri, ***"An In-Context Learning Agent for Formal Theorem-Proving,"*** First Conference on Language Modeling (COLM), 2024. [arXiv][Code]

A. Gautam, **A. Thakur**, G. Dhanania and S. Mohan, ***"A distributed algorithm for balanced multi-robot task allocation,"*** 2016 11th International Conference on Industrial and Information Systems (ICIIS), Roorkee, 2016, pp. 622-627, doi: 10.1109/ICIINFS.2016.8263014. [IEEE Link]

Invited Talks

Amazon's Automated Reasoning Symposium,***"End-to-End Verified Code Generation in Lean,"*** Oct 2025. [Talk Details/Abstract]

TPC 25 (Trillion Parameter Consortium), ***"CLEVER: A Benchmark for Two-staged, End-to-end Verified Code Generation"*** Jul 2025 [Talk Details/Abstract]

Special Seminar at Department of Mathematics at Rutgers, ***"Advancing Theorem Proving with AI: The Power of Large Language Models,"*** Feb 2024. [Talk Details/Abstract]

Awards

2024: Amazon Automated Reasoning Fellowship UT-Amazon Science Hub

Experience

Applied Science Intern May 2026 – Aug 2026
Amazon Science (Automated Reasoning Group) *Austin, USA*

- Working on AI-agent-based proof automation for the Strata team at Amazon, applying automated reasoning and formal methods to software verification.

Machine Learning Intern May 2025 – Aug 2025
Asari AI *San Francisco, USA*

- Building AI agents and frameworks for complex systems design and reasoning.

Software Engineer II ***(Software Engineer, 2017–2019)*** July 2017 – July 2022
Microsoft India Development Center *Hyderabad, India*
Microsoft Azure Backup (MAB) team

- Designed a backup micro-service for hierarchical cloud objects (e.g. File Shares), with a parallel file-system diffing algorithm for incremental backups, **proved correct** and specified in **TLA+**.
- Built a distributed, map-reduce-style billing orchestrator for Azure Backup, making billing computation **6x** faster.
- Built unsupervised log mining (clustering + decision-tree explanations) to infer root causes of customer issues; adopted by service engineers. Mentored interns.

Research Intern Jan 2017 - June 2017
Microsoft Research *Bangalore, India*

- Program synthesis with the *PROSE (Program Synthesis using Examples) SDK* team: synthesizing programs over a restricted grammar from input-output examples, and using ML to rank candidates and pick the most useful one.

Software Engineering Intern May 2016 - July 2016
Microsoft India Development Center *Hyderabad, India*

- Cut recovery time for parts of large ($\geq$ 1 TB) backed-up disks from **10 hrs to 3 mins** using iSCSI and mocked NTFS layers.

Summer Research Fellow May 2015 - July 2015
National Institute of Science Education & Research *Bhubaneswar, India*

- Computational number theory with Prof. Brundaban Sahu: time complexity of Lenstra's Elliptic Curve Factorization and its relation to y-smooth numbers.

Project Trainee May 2014 - July 2014
Bhabha Atomic Research Centre (BARC) *Mumbai, India*

- Worked in development of Network Management System and automation modules.

Other Skills

Formal Methods & PL: Lean 4, Coq, TLA+, SMT solvers, automated theorem proving, DSL design
Machine Learning: deep learning, LLM post-training (SFT, RL), LLM agents, PyTorch, Ray
Systems: distributed system design, concurrency & locking, file systems, map-reduce-style parallelism
Languages: Python, Lean, Rocq, Rust, C#, C/C++, Java, JavaScript